

IL'INSKOYE, Russian Federation

WMO: 321210

Lat: 47.9867N

Lon: 142.2050E

Elev: 56

StdP: 14.67

Time Zone: 11.00 (E11)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.0	-1.7	-12.1	2.9	-4.4	-8.3	3.5	-0.3	33.6	18.1	31.0	16.4	4.4	110	0.713

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.0	72.0	67.7	70.1	66.1	68.4	64.8	68.6	70.9	67.0	69.4	65.4	67.9	10.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.5	101.3	70.2	65.9	95.8	68.9	64.2	90.2	67.4	33.0	71.3	31.7	69.5	30.4	67.9	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
30.2	26.2	22.9	-15.6	75.5	3.4	2.3	-18.1	77.2	-20.1	78.5	-22.0	79.9	-24.5	81.5		
			-15.9	70.2	3.4	1.9	-18.3	71.6	-20.3	72.7	-22.2	73.8	-24.6	75.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.6	12.5	13.3	23.4	34.4	43.1	51.4	59.1	62.7	57.0	45.0	30.4	17.7
	DBStd	18.44	7.73	8.04	6.93	5.06	5.50	4.82	4.46	4.34	5.16	6.07	7.94	7.52
	HDD50	5523	1161	1028	823	468	225	40	1	0	7	179	588	1003
	HDD65	10023	1626	1448	1288	917	678	409	188	99	243	621	1038	1468
	CDD50	1006	0	0	0	0	11	81	282	393	216	23	0	0
	CDD65	33	0	0	0	0	0	0	4	27	2	0	0	0
	CDH74	10	0	0	0	0	0	1	2	7	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	9.9	10.3	9.7	10.3	10.0	9.9	8.4	8.1	8.2	9.3	11.1	12.1	11.2
Precipitation	PrecAvg	32.6	2.2	1.7	1.8	2.0	2.5	2.1	3.3	4.0	3.6	3.7	2.9	2.8
	PrecMax	46.1	4.1	3.3	4.1	4.5	4.8	4.3	10.7	8.3	6.8	7.5	6.2	5.1
	PrecMin	25.0	0.7	0.1	0.7	0.6	0.6	0.1	0.2	1.1	1.3	1.7	0.8	0.6
	PrecStd	5.4	0.7	0.7	1.0	1.0	1.0	1.1	2.5	1.7	1.3	1.3	1.4	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.9	35.0	40.3	52.6	61.5	68.9	72.0	74.9	71.9	63.0	51.6	38.8
		MCWB	31.4	32.9	36.6	44.8	53.0	61.2	66.2	70.2	66.6	58.3	49.1	37.6
	2%	DB	29.9	31.5	37.4	47.9	57.5	63.2	69.5	72.9	69.2	59.7	47.9	35.1
		MCWB	27.7	29.5	34.2	42.4	50.8	57.8	65.2	69.0	64.7	55.3	45.2	33.1
	5%	DB	27.2	28.7	35.1	45.2	54.6	60.8	67.8	71.4	67.4	57.2	44.8	31.9
		MCWB	25.2	27.0	32.5	40.8	49.1	56.8	64.2	67.7	63.2	53.2	42.1	29.9
	10%	DB	24.4	25.7	33.1	42.6	52.2	58.7	66.2	69.8	65.6	54.7	41.8	29.3
		MCWB	22.5	23.5	30.6	38.8	47.7	55.2	63.2	66.2	61.4	50.6	39.1	27.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.5	33.5	37.6	45.9	54.5	61.8	67.8	71.8	67.8	59.0	49.4	37.8
		MCDB	32.4	34.9	39.5	50.3	59.1	67.7	70.1	73.7	70.8	61.7	51.4	38.7
	2%	WB	27.9	29.9	34.7	43.2	51.9	59.0	66.3	69.9	65.5	56.0	45.4	33.3
		MCDB	29.5	31.2	36.6	46.9	56.1	62.0	68.6	72.2	68.4	59.1	47.6	34.7
	5%	WB	25.4	27.0	32.8	41.2	50.0	57.4	64.9	68.3	63.8	53.7	42.3	30.1
		MCDB	27.1	28.8	35.0	44.7	53.8	60.2	67.2	70.6	66.8	56.5	44.5	31.6
	10%	WB	22.6	23.6	30.8	39.1	48.1	55.6	63.5	67.0	61.9	51.3	39.5	27.3
		MCDB	24.3	25.5	32.9	42.2	51.6	58.1	65.9	69.4	65.0	54.3	41.8	29.2
Mean Daily Temperature Range	5% DB	MDBR	11.4	13.1	10.8	9.1	9.8	9.3	8.5	9.0	11.6	11.4	9.2	10.2
		MCDBR	12.1	13.6	12.1	13.4	13.5	13.2	10.8	10.3	13.3	13.6	11.7	11.7
		MCWBR	11.4	12.5	10.5	9.8	9.0	8.9	7.6	7.7	10.3	11.6	11.1	11.4
	5% WB	MCDBR	11.5	12.4	11.4	12.5	11.8	11.7	9.5	8.6	11.9	13.0	11.7	11.7
		MCWBR	11.2	11.9	10.4	9.9	8.5	8.4	7.4	6.9	10.1	11.8	11.4	11.8
Clear-Sky Solar Irradiance	taub	0.316	0.353	0.412	0.498	0.531	0.501	0.464	0.440	0.397	0.404	0.377	0.338	
	taud	2.099	1.974	1.892	1.812	1.840	2.036	2.219	2.298	2.344	2.145	2.102	2.083	
	Ebn at Noon	227	243	247	238	236	246	253	255	255	225	200	199	
	Edh at Noon	28	41	52	62	63	52	43	38	33	34	29	26	
All-Sky Solar Radiation	RadAvg	406	710	1086	1404	1513	1572	1421	1345	1182	775	427	298	
	RadStd	46	62	105	152	144	162	158	126	112	51	38	30	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page